



[HINT SOLUTION]

Q.1 10

Sol. Given, $z^6 + z^4 - 2 = 0$

Put $z^2 = t$

$$\Rightarrow t^3 + t^2 - 2 \Rightarrow (t-1)(t^2 + 2t + 2) = 0$$

$$\therefore t = 1, -1 + i, -1 - i$$

$$\text{i.e., } z^2 = 1 \text{ or } z^2 = -1 + i \text{ or } z^2 = -1 - i$$

$$\therefore |z|^4 = 1, |z|^4 = 2 \text{ or } |z|^4 = 2$$

$$\text{Hence, } \sum_{i=1}^6 |z_i|^4 = 2(1 + 2 + 2) = 10.$$

Q.2 68

Sol. Given, $|z - 2i| \leq 1$

$$\therefore m = 2 \left(\sqrt{\frac{1}{4} + 4} - 1 \right) = 2 \left(\frac{\sqrt{17}}{2} - 1 \right)$$

$$\text{and } M = 2 \left(\sqrt{\frac{1}{4} + 4} + 1 \right) = 2 \left(\frac{\sqrt{17}}{2} + 1 \right)$$

$$\therefore |2z + 1| = 2 \left| z + \frac{1}{2} \right|$$

$$= 2 \left| z - \left(-\frac{1}{2} + i \cdot 0 \right) \right|$$

$$= \left(\underbrace{(\sqrt{17} - 2)}_m + \underbrace{(\sqrt{17} + 2)}_M \right)$$

$$\text{So, } (m + M)^2 = (2\sqrt{17})^2 = 4 \times 17 = 68.$$

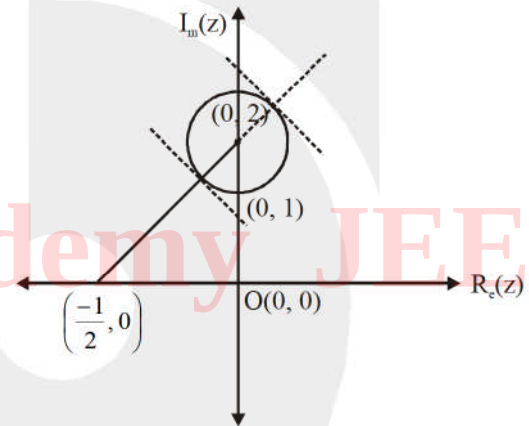
Q.3 5

Sol. Given, $z^3 + \frac{3(\bar{z})^2}{|z|} = 0$ (1)

Let $z = re^{i\theta}$ be the solution of equation (1), where $|z| = r (\neq 0)$ and $\theta = \arg(z)$.

$$\text{So, put } z = re^{i\theta} \text{ in equation (1), we get } r^3 e^{i3\theta} + \frac{3r^2 e^{i(-2\theta)}}{r} = 0.$$

$$\Rightarrow r^3 e^{i3\theta} + \frac{3r}{e^{i2\theta}} = 0 \quad (\text{As, } r \neq 0)$$



$$\Rightarrow r^2 e^{i5\theta} + 3 = 0 \Rightarrow r^2 e^{i5\theta} = -3 \quad \dots\dots(2)$$

$$\therefore |r^2 e^{i5\theta}| = |-3| \Rightarrow r^2 = 3$$

$$\therefore r = \sqrt{3}$$

$\Rightarrow$ All solutions of given equation lies on circle centred at origin and radius $\sqrt{3}$ i.e., $|z| = \sqrt{3}$.

$\therefore$ On putting $r = \sqrt{3}$ in equation (2), we get $e^{i5\theta} = -1$

$$\Rightarrow \cos 5\theta + i \sin 5\theta = -1 + i \cdot 0$$

$$\therefore \cos 5\theta = -1, \theta \in [0, 2\pi]$$

$$\Rightarrow 5\theta = \pi, 3\pi, 5\pi, 7\pi, 9\pi$$

$$\therefore \theta = \frac{\pi}{5}, \frac{3\pi}{5}, \pi, \frac{7\pi}{5}, \frac{9\pi}{5}$$

So, the given equation will have 5 solutions. **Ans.**

Aliter: Given that, $z^3 + \frac{3(\bar{z})^2}{|z|} = 0 \quad \dots\dots(1)$

As, $|z|^2 = z\bar{z} \Rightarrow \frac{|z|^4}{z^2} = (\bar{z})^2 \quad \dots\dots(2)$

$\therefore$ Using (2) in (1), we get $z^3 + \frac{3}{|z|} \left(\frac{|z|^4}{z^2} \right) = 0$

$$\Rightarrow z^3 + \frac{3|z|^3}{z^2} = 0 \Rightarrow z^5 + 3|z|^3 = 0$$

As, $|z| \neq 0$

$\therefore$ Put $3|z|^3 = \lambda$, where λ is positive real number.

So, we get $z^5 + \lambda = 0 \Rightarrow z = (-\lambda)^{1/5}$

$\therefore$ The above equation will have 5 distinct solution.

Q.4 $\frac{25}{4}$

Sol. Given, $|z_1 + iz_2| = |z_1| + |z_2|$

$$\Rightarrow |z_1 + iz_2| = |z_1| + |z_2|$$

$\therefore z_1, iz_2, 0 + i0$ are collinear and (z_1) and (iz_2) are on same side of origin also.

$$\therefore \arg(z_1) = \arg(iz_2)$$

$$\Rightarrow \arg(z_1) = \frac{\pi}{2} + \arg(z_2)$$

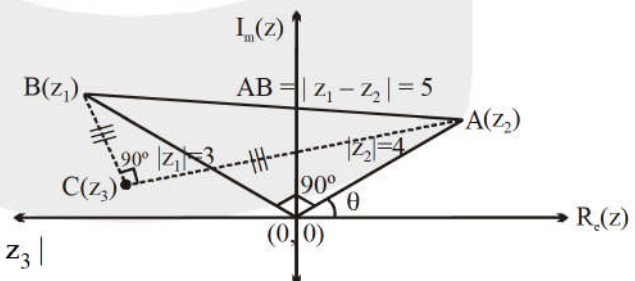
$$\text{Let } z_3 = \frac{z_2 - iz_1}{1 - i} \Rightarrow (1 - i)z_3 = (z_2 - iz_1)$$

$$\Rightarrow (z_2 - z_3) = i(z_1 - z_3) \Rightarrow |z_2 - z_3| = |z_1 - z_3|$$

$$\therefore BC = AC$$

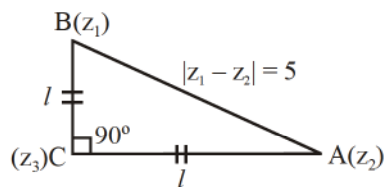
$$\text{Also, } \left(\frac{z_2 - z_3}{z_1 - z_3} \right) = e^{\frac{i\pi}{2}}$$

$$\Rightarrow \angle BCA = 90^\circ \text{ (using rotation)}$$



As, $l^2 + l^2 = 25 \Rightarrow l^2 = \frac{25}{2}$

So, $\Delta = \frac{1}{2} l^2 = \frac{1}{2} \left(\frac{25}{2} \right) = \frac{25}{4}$. **Ans.**



Q.5 2

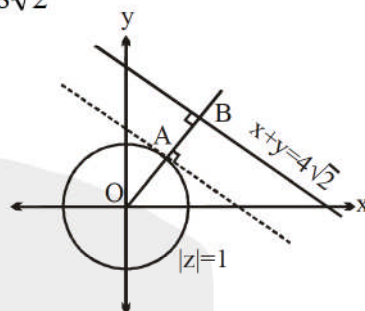
Sol. $|z_1| = 2$, a circle of radius 2 and $(1-i)z_2 + (1+i)\bar{z}_2 = 8\sqrt{2}$

$\Rightarrow$ a straight line $x+y = 4\sqrt{2}$

$\therefore$ AB is minimum along a line $y=x$

$A = (\sqrt{2}, \sqrt{2})$, $B = (2\sqrt{2}, 2\sqrt{2})$

$\therefore AB = \sqrt{(2\sqrt{2} - \sqrt{2})^2 + (2\sqrt{2} - \sqrt{2})^2}$
 $= \sqrt{2+2} = \sqrt{4} = 2$



Q.6 $\frac{-1+i}{2}$

Sol. The given point $z_1 = \frac{2-i}{3+i} \times \frac{3-i}{3-i} = \frac{5-5i}{10} = \left(\frac{1}{2}, \frac{-1}{2} \right)$

The given straight line is $(1+i)z + (1-i)\bar{z} = 0$

Put $z = x + iy$, we get

$(1+i)(x+iy) + (1-i)(x-iy) = 0$

$\Rightarrow 2x - 2y = 0 \Rightarrow y = x$

So, the image of point $\left(\frac{1}{2}, \frac{-1}{2} \right)$ in the line $y = x$ is $\left(\frac{-1}{2}, \frac{1}{2} \right) = \frac{-1+i}{2}$.

Q.7 25

Sol. $z^4 + 3z^2 + 1 = (z-z_1)(z-z_2)(z-z_3)(z-z_4)$

Putting $z = 2i$ & $-2i$

$16 - 12 + 1 = (2i - z_1)(2i - z_2)(2i - z_3)(2i - z_4)$ (1)

$16 - 12 + 1 = (-2i - z_1)(-2i - z_2)(-2i - z_3)(-2i - z_4)$ (2)

Multiplying (1) & (2)

$(z_1^2 + 4)(z_2^2 + 4)(z_3^2 + 4)(z_4^2 + 4) = 25$. **Ans.**

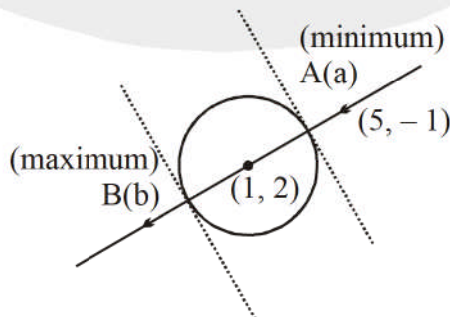
Q.8 0

Sol. Given that,

$|z - (1 + 2i)| = r, 0 < r < 5$

$|z - 5 + i| = |z - (5 - i)|$

Now, $\arg \left(\frac{a-5+i}{b-1-2i} \right)$



$$= \arg(a - (5-i)) - \arg(b - (5-i))$$

[Using vectorial representation of complex number]

$$= 0. \text{ Ans.}$$

Q.9 $\left(-\frac{\pi}{4}, 0\right)$

Sol. As, $\frac{kz}{k+1}$ represents any point lying on the joining origin and z .

Given, $\left|\frac{kz}{k+1} + 2i\right| > \sqrt{2}$

Hence, $\frac{kz}{k+1}$ should lie outside the circle $|z + 2i| = \sqrt{2}$.

So, z should lie in the shaded region.

$$\therefore -\frac{\pi}{4} < \arg(z) < 0. \text{ Ans.}$$

Aliter:

Put $z = x + iy$

$$\left|\frac{k(x+iy)}{k+1} + 2i\right| > \sqrt{2}$$

$$\Rightarrow x^2 + y^2 + 4\left(\frac{k+1}{k}\right)y + 2\left(\frac{k+1}{k}\right)^2 > 0$$

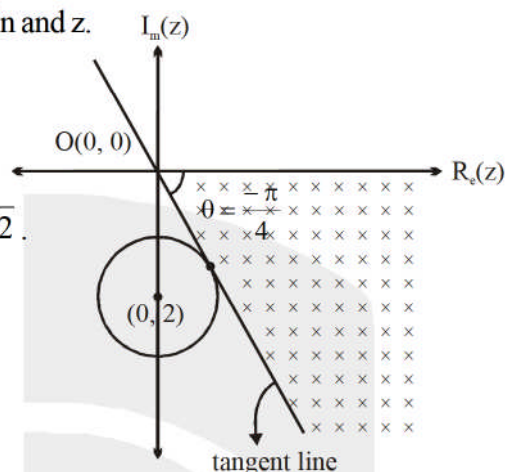
If $k \in \mathbb{R} \Rightarrow \frac{k+1}{k} \in \mathbb{R}$

$\therefore$ Quadratic in $\frac{k+1}{k}$ will have $D < 0$

$$\Rightarrow 16y^2 - 8(x^2 + y^2) < 0 \Rightarrow y^2 - x^2 < 0 \Rightarrow \left|\frac{y}{x}\right| < 1$$

$$\Rightarrow -1 < \frac{y}{x} < 0 \quad [\text{As in fourth quadrant } x > 0, y < 0.]$$

$$\Rightarrow -1 < \tan \theta < 0 \Rightarrow \theta \in \left(-\frac{\pi}{4}, 0\right).$$

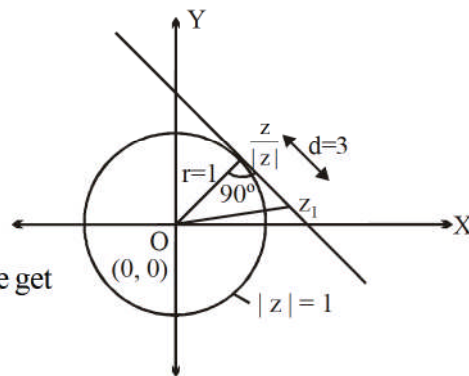


Q.10 $\sqrt{10}$

Sol.

Note that: $\left|\frac{z}{|z|}\right| = 1 \Rightarrow \frac{z}{|z|} = 1$ is unimodular complex number.

Given, $\arg \left(\frac{z_1 - \frac{z}{|z|}}{\frac{z}{|z|}} \right) = \frac{\pi}{2}$ and $\left| \frac{z}{|z|} - z_1 \right| = 3$



$\therefore$ From above diagram, using Pythagoras theorem, we get

$$|z_1| = \sqrt{(1)^2 + (3)^2} = \sqrt{10} \text{ . Ans.}$$

Aliter: As $\arg \left(\frac{z_1 - \frac{z}{|z|}}{\frac{z}{|z|}} \right) = \frac{\pi}{2} \Rightarrow \left(\frac{z_1 - \frac{z}{|z|}}{\frac{z}{|z|}} \right) = i\lambda \ (\lambda \in \mathbb{R})$ [purely imaginary]

Also, $\left| \frac{z_1 - \frac{z}{|z|}}{\frac{z}{|z|}} \right| = |i\lambda| \Rightarrow |\lambda| = 3$

$\therefore \lambda = \pm 3$

So, $\left(z_1 - \frac{z}{|z|} \right) = \pm 3i \left(\frac{z}{|z|} \right)$

$\Rightarrow z_1 = (1 \pm 3i) \frac{z}{|z|}$

So, $|z_1| = \left| (1 \pm 3i) \frac{z}{|z|} \right| \Rightarrow |z_1| = \sqrt{10} \text{ . Ans.}$

Q.11 2

Q.12 8

Q.13 3

Sol. Given, $(z + \bar{\omega}) = z^2$ (1)

and $\omega + \bar{z} = \omega^2$ (2)

$\therefore$ Taking conjugate on both sides of equation (2), we get

$(\bar{\omega} + z) = \bar{\omega}^2$ (3)

$\therefore$ From equations (1) and (3), we get

$z^2 = \bar{\omega}^2 \Rightarrow (z - \bar{\omega})(z + \bar{\omega}) = 0$

$\therefore$ Either $z = \bar{\omega}$ or $z = -\bar{\omega}$

Case-I: When $\bar{\omega} = z$

$\therefore$ Equation (1) is $2z = z^2 \Rightarrow z(z - 2) = 0$

$\therefore z = 0, 2$

So, $(z, \omega) = \begin{cases} (z = 0, \omega = 0) \\ (z = 2, \omega = 2) \end{cases}$

Case-II: When $\bar{\omega} = -z$

$\therefore$ Equation (1) is $z - z = z^2 \Rightarrow z^2 = 0$

$\therefore z = 0$

So, $(z, \omega) = (z = 0, \omega = 0)$

(i) Number of ordered pairs (z, ω) are two i.e. $(z = 0, \omega = 0)$ or $(z = 2, \omega = 2)$.

(ii) $\text{Sum} = \sum z_r^2 + \sum \omega_r^2 = ((0)^2 + (2)^2) + ((0)^2 + (2)^2) = 4 + 4 = 8$. **Ans.**

(iii) The sum of an infinite G.P.,

$$|z| + \frac{|z|}{|\omega+1|} + \frac{|z|}{|\omega+1|^2} + \frac{|z|}{|\omega+1|^3} + \dots \infty$$

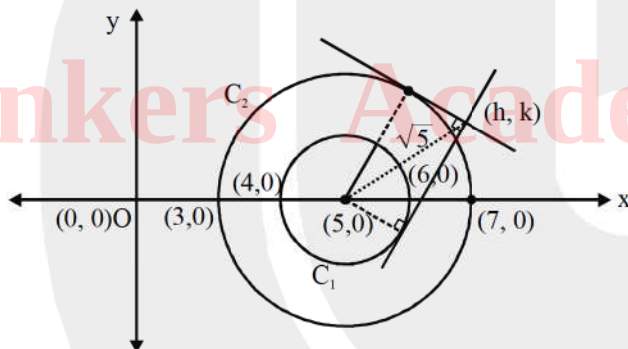
$$= 2 + \frac{2}{3^1} + \frac{2}{3^2} + \frac{2}{3^3} + \dots \infty = 2 \left(\frac{1}{1 - \frac{1}{3}} \right) = \left(\frac{1}{\frac{2}{3}} \right) = 3. \text{ Ans.}$$

Q.14 $|z - 5| = \sqrt{5}$

Q.15 1

Sol. $C_1: x^2 + y^2 - 10x + 24 = 0$ whose centre is $(5, 0)$ and radius is 1.

$C_2: x^2 + y^2 - 10x + 21 = 0$ whose centre is $(5, 0)$ and radius is 2.



Q.16 0004

Sol. Given, $(1 + \omega^2)^8 = a + b\omega$ (1)

($a, b \in \mathbb{R}$ and ω be any non-real cube root of unity.)

As, $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$ (2)

$$\Rightarrow (1 + \omega^2) = -\omega$$

So, equation (1) becomes

$$(-\omega)^8 = a + b\omega$$

$$\Rightarrow \omega^8 = a + b\omega \Rightarrow \omega^2 = a + b\omega \text{ (As, } \omega^3 = 1)$$

$$\Rightarrow \boxed{-1 - \omega = a + b\omega} \quad [\text{As, } \omega^2 = -1 - \omega]$$

So, on comparing we get

$$a = 1, b = -1$$

$$\text{Hence, } |(2a^{-1} + 2b^{-1})| = |-2 - 2| = 4.$$

Q.17 0002

Sol. Given, $|z_1| = |z_2| = |z_3| = 1$

$$\Rightarrow z_1 \bar{z}_1 = 1 \Rightarrow \bar{z}_1 = \frac{1}{z_1}$$

$$\text{As, } (z_1 + z_2 + z_3)^2 = (z_1^2 + z_2^2 + z_3^2) + 2(z_1 z_2 + z_2 z_3 + z_3 z_1)$$

$$\text{But } z_1^2 + z_2^2 + z_3^2 = 0 \text{ (given)}$$

$$\Rightarrow (z_1 + z_2 + z_3)^2 = 2z_1 z_2 z_3 \left(\frac{1}{z_3} + \frac{1}{z_1} + \frac{1}{z_2} \right)$$

$$\Rightarrow |z_1 + z_2 + z_3|^2 = 2 |z_1 z_2 z_3| | \bar{z}_3 + \bar{z}_1 + \bar{z}_2 |$$

$$\Rightarrow |z_1 + z_2 + z_3|^2 = 2 |z_1 + z_2 + z_3|$$

$$\Rightarrow |z_1 + z_2 + z_3| = 0 \text{ or } 2$$

$$\text{As, } z_1 + z_2 + z_3 \neq 0, \text{ so } |z_1 + z_2 + z_3| \neq 0.$$

$$\text{Hence, } |z_1 + z_2 + z_3| = 2. \text{ Ans.}$$

Q.18 0100

$$\text{Sol. } (z - (3 + 4i))(\bar{z} - (3 - 4i)) = 25$$

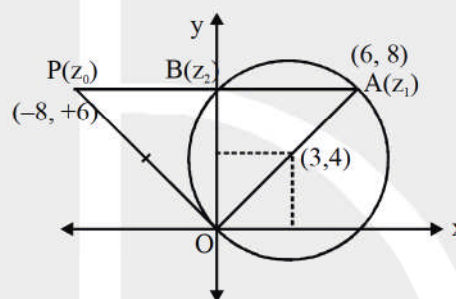
$$|z - (3 + 4i)| = 5$$

$$z_1 = 6 + 8i$$

$$z_0 = (6 + 8i)i = -8 + 6i$$

$$PA \cdot PB = OP^2$$

$$|z_0 - z_1| \cdot |z_0 - z_2| = 100$$



Q.19 0006

$$\text{Sol. Consider } (a + b\omega + c\omega^2) = a + b \left(\frac{-1 + i\sqrt{3}}{2} \right) + c \left(\frac{-1 - i\sqrt{3}}{2} \right)$$

$$= \left(a - \frac{b}{2} - \frac{c}{2} \right) + \frac{i\sqrt{3}}{2} (b - c)$$

$$\text{So, } |a + b\omega + c\omega^2|^2 = \left(a - \frac{b}{2} - \frac{c}{2} \right)^2 + \frac{3}{4} (b - c)^2 = \frac{(2a - b - c)^2 + 3(b - c)^2}{4}$$

$$= \frac{(4a^2 + b^2 + c^2 - 4ab + 2bc - 4ac) + 3(b^2 + c^2 - 2bc)}{4} = \frac{4a^2 + 4b^2 + 4c^2 - 4ab - 4bc - 4ca}{4}$$

$$= (a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\therefore |a + b\omega + c\omega^2|^2 = \frac{(a - b)^2 + (b - c)^2 + (c - a)^2}{2}$$

$$\text{Hence, } \left[\frac{|a + b\omega + c\omega^2|^2}{2} \right]_{\text{min. value}} = \left[\frac{(a - b)^2 + (b - c)^2 + (c - a)^2}{4} \right]_{\text{Take } a=1, b=3 \text{ \& } c=5}$$

$$= \frac{4 + 4 + 16}{4} = \frac{24}{4} = 6. \text{ Ans.}$$



Q.20 7

Sol. $|z + \bar{z}| + 2|z - \bar{z}| = 4$

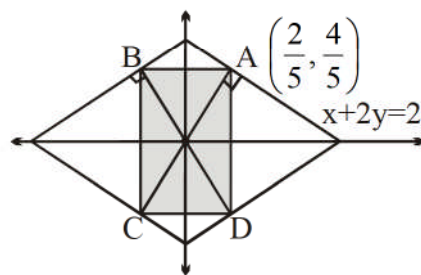
$$\Rightarrow |2x| + 2|2iy| = 4$$

$$\Rightarrow |x| + 2|y| = 2$$

[Note: $|z|$ is minimum when it is perpendicular.]

Required area = Area of rectangle

$$= \frac{4}{5} \times \frac{8}{5} = \frac{32}{25} \text{ sq. units.}$$



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